Agentic AI Bootcamp

Applications: Teaching & Research

Erkmen G. Aslim & **Emily Beam**

Department of Economics, University of Vermont

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thinkingwithagents.github.io

Where We Left Off & Today's Plan

From Session 1

- Agentic AI vs. chat AI
- Context window & planning
- `CLAUDE.md`, `PLAN.md`, `SKILL.md`
- Permissions & safety
- App 1 (live): website redesign
- App 2 (designed): **lecture-builder**

Today: Three Live Applications

1. **Lecture Builder** — create the skill we designed last time, then run it on real papers
2. **Research Brainstorm** — stress-test an audience question live
3. **Web Scraper** — novel data on screen in 10 minutes

Each app is end-to-end: prompt → skill → output. Prompts are on the slides — copy as we go.

Setup: Install Your Skill Toolkit

Anthropic skills (one-time)

1. **Add the marketplace:**

```
/plugin marketplace add anthropics/skills
```

2. **Install the document-skills bundle:**

```
/plugin install
  document-skills@anthropic-agent-skills
```

Gives you: `pdf`, `docx`, `pptx`, `xlsx`, `frontend-design`, `skill-creator`, `algorithmic-art`.

Third-party skills via `npx`

Full economist pack:

```
npx skills add thinkingwithagents/skills
```

One skill at a time:

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Includes: `research-brainstorm`, `find-data`, `academic-beamer-deck`.

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! `/skill-creator` is **not** bundled with Claude Code by default — it ships inside `document-skills`. Install it before App 1.

Application 1

Lecture Builder Skill — From Papers to a Lecture

Where We Left Off — The Skill Spec

```
/skill-creator

Create a skill called "lecture-builder" that converts academic documents into lecture materials.  The skill should:

1.  Read documents in their entirety (PDF, DOCX, web pages) -- break reading into parts to avoid excessive token

usage.  Include a Python parsing script.

2.  Extract:  key arguments, empirical findings, methodological details, discussion points, potential exam questions.

3.  Generate two outputs:

a) Structured lecture notes (markdown) with section headers, key takeaways, discussion prompts

b) A polished slide deck (draw on \academic-beamer-deck or \pptx) with clean visuals, one idea per slide

4.  Adapt tone for undergraduate vs. graduate audiences (configurable).

Save SKILL.md + supporting files to .claude/skills/lecture-builder/.
```

Last week we wrote the spec. ***Today we actually build the skill — then run it.***

Building the Skill: `/skill-creator` Walkthrough

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`/skill-creator` **interviews you**. Be ready to answer:
name, audience, inputs, outputs, edge cases, where to save.

Anatomy of `lecture-builder`

`.claude/skills/lecture-builder/`

- `SKILL.md` playbook
- `parse_document.py` PDF/DOCX reader
- `slide_template.tex` UVM Beamer
- `notes_template.md` notes scaffold

What `SKILL.md` enforces

- Chunked reading → token control
- Mandatory citations from source PDFs
- Undergrad vs. grad tone toggle
- Dual output: notes + slides
- Reuses UVM color palette

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Skills with Python scripts > pure-markdown skills when tokens matter. The script reads; the LLM synthesizes.

The Input Folder

`./inputs/`

- `gov_thinktank_ice_arrests.pdf` — policy report
- `ice_enforcement_report.pdf` — government brief
- `aslim_paper.pdf` — academic working paper

Why this mix?

- Heterogeneous sources: **policy** + **government** + **academic**
- Forces the skill to harmonize tone, terminology, and evidence quality
- Mirrors how you'd actually build a lecture

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Topic: The labor and health consequences of ICE enforcement. Connects directly to ongoing research at UVM.

The Execution Prompt

```
/lecture-builder

I have a folder ./inputs/ with:  gov_thinktank_ice_arrests.pdf,
ice_enforcement_report.pdf, aslim_paper.pdf.

Build a 50-minute undergraduate lecture on the labor and health
consequences of ICE enforcement.  Audience:  junior/senior econ majors
who have taken intermediate micro and one applied econometrics class.

Outputs to ./outputs/:
  1) lecture_notes.md -- sectioned notes with 5 key takeaways,
     4 discussion prompts, 3 candidate exam questions.
  2) lecture_slides.tex -- Beamer deck (~20 slides, 16:9).  Use
     a clean academic palette -- default UVM (deep green / gold) or ask
     me.  In-text citations to all three source PDFs.

Be conservative with tokens:  parse each PDF in chunks via
parse_document.py, summarize per-section, then synthesize.
```

Audience spec + output spec + token discipline. Skills work better when the prompt is concrete.

Expected Output (Live Demo Running)

▶ Live demo running in the background — final files appear by the end of this section

lecture_notes.md

 # 1. The setting
Recent expansions of interior immigration enforcement...
 # 2. Identification strategy
287(g) rollout as a staggered DiD...
 # 3. Findings
Aslim et al. document...
 # Discussion prompts
1. Why might the effect on infant health attenuate over time?
 # Exam questions
1. Sketch the parallel-trends test...

synthesizes →

lecture_slides.tex

```
% 20 slides, 16:9
% UVM palette
\begin{frame}{Setting}
  ICE arrests timeline
\end{frame}
\begin{frame}{Methods}
  staggered DiD
\end{frame}
\begin{frame}{Results}
  main effect
\end{frame}
```

What Just Happened (Under the Hood)

Read in chunks

parse_document.py

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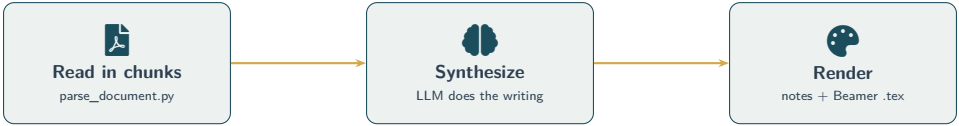
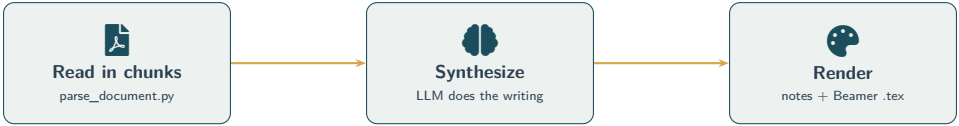


What Just Happened (Under the Hood)



This skill is reusable. Swap the input folder, swap the topic, swap the audience — get a new lecture in minutes. Build once, deploy forever.

Application 2

`/research-brainstorm` — Stress-Test a Research Idea

What the Research Brief Contains

Brief sections

- **The question** — structured: unit, population, treatment, outcome, comparison, setting
- **Closest related work** — 3–6 cites with sharp delta
- **Contribution type** — setting / identification / reframing / measurement / mechanism / policy
- **The strongest objection** — one paragraph, unvarnished
- **Alternative framings** — 2–3 pivots
- **Feasibility** — data, identification, power, timeline, fatal risks
- **Next concrete step**

Saved to:

`research_briefs/YYYY-MM-DD-slug.md`

Markdown so it lives next to your other notes. Re-openable, editable, citeable in your own write-ups.

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The conversation is great.
The artifact is what you keep.

Installing /research-brainstorm

Full economist pack

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```

Installs the full pack in one go. Today we use: `research-brainstorm`, `find-data`, `academic-beamer-deck`.

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Mix-and-match per project. Lighter footprint when you only need one.

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Same `npx skills` mechanism works for any GitHub-hosted skill repo. **Build your own pack and share it with your coauthors.**

Live Demo: Hand Me a Question

Audience:

hand me a half-formed research question.

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An effect
you suspect
but can't yet measure

A dataset
you have access to
but haven't used

A policy
you want
to evaluate

Live: invoke `/research-brainstorm`, ride through the phases, walk away with a brief.

Application 3

Web Scraper — From `/find-data` to a Live Visualization

`/find-data` Picks NBER Working Papers

`/find-data` in one line

We hand it our constraints (applied-micro audience, 24+ months, scrape-able, time-series + cross-section). It scans repositories and surfaces a source.

Today's pick: NBER working papers.

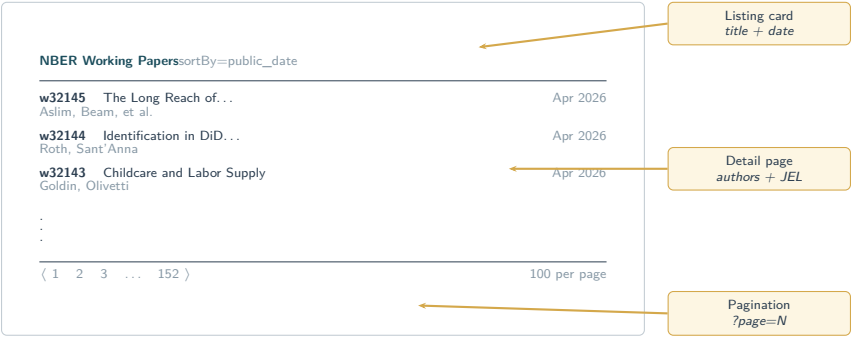
Why NBER: every economist cares; clean HTML; rich JEL + affiliation cross-section; ~1,500 papers / year.

URL:
`nber.org/papers?page=N&perPage=100&sortBy=public_date`

Extract: `nber_number, title, authors, public_date, primary_jel_code, affiliations.`

Goal: novel data on the screen in under 10 minutes. Skip the buildup — go straight to scraping.

Inspect Before You Prompt



View-source first. The agent writes better scrapers when you've already named the selectors.

The Scraping Prompt

```
Scrape NBER working papers for a teaching demo.  Build a self-contained

Python script in ./scrape_nber/ using uv:

1. Fetch the most recent ~1,000 papers from

   nber.org/papers?page=N&perPage=100&sortBy=public_date

   (paginate until 1,000 or cutoff 2024-01-01).

2. Extract:  nber_number, title, authors, public_date, primary

   JEL code, affiliations (from detail page if needed).

3. Save to ./scrape_nber/data/nber_papers.csv.

4. Generate three PNGs to ./scrape_nber/figures/:

   - papers_per_month.png   bars, last 24 months, UVM colors

   - top_jel_codes.png   top 15 primary JEL codes

   - top_institutions.png   top 10 affiliations

5. Print 5-line stdout summary.

Politeness:  real-browser User-Agent; sleep 0.5s between fetches;

cache HTML to ./scrape_nber/cache/ so re-runs are free.
```

What Claude Does While We Keep Talking

Fetching pages...

Parsing...

Plotting...

```
$ uv run scrape_nber/main.py
[INFO] fetching page 1/10 ...ok (98 papers)
[INFO] fetching page 2/10 ...ok (100 papers)
[INFO] ...
[INFO] parsing JEL codes ...
[INFO] saved data/nber_papers.csv
[INFO] writing figures/...

== Summary ==
n_papers:  1,000
date range:  2024-04 to 2026-04
top JEL: J (Labor)
top affil:  NBER, Harvard, MIT
```

The terminal is a TV. Let it run while you talk through what's coming.

What Claude Does While We Keep Talking

✔ Fetched 10 / 10 pages

Parsing 1,000 records. . .

Plotting. . .

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◌ **Plotting 3 figures. . .**

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What Claude Does While We Keep Talking

- ✔ Fetched 10 / 10 pages
- ✔ Parsed 1,000 records
- ✔ Wrote 3 PNGs

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Three workflows. Three artifacts on disk. **All reusable.**

Thinking With Agents

Online sessions coming this summer

Workshops, deep-dives, and Q&A on agentic AI for academic research — built by economists, for economists and academics alike.

Stay tuned for dates and registration:

`thinkingwithagents.github.io`